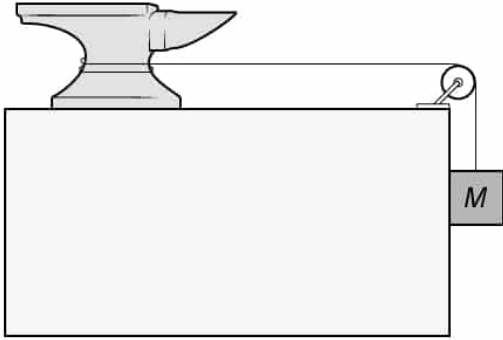


Subject:
Physics

System:
Mechanics and Energy

A cable with a tension of 45 N is used to suspend a 5 kg mass M at rest against a wall. A horizontal force (not shown) also pushes the mass against the wall.



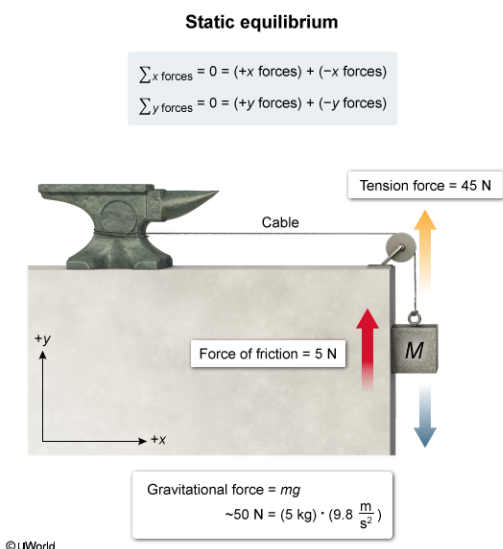
What is the magnitude and direction of the force of friction between the mass and the wall?

- ✔ ☒ A. 5 N, upward [70%]
☐ B. 5 N, downward [15%]
☐ C. 15 N, upward [10%]
☐ D. 15 N, downward [4%]

Correct

69% answered correctly

Explanation:



Any object may be acted on by multiple **translational forces** of different origin, magnitude, and direction. For example, an object located in two-dimensional space will **accelerate** when the sum of all forces acting along one or both axes of the xy-coordinate system does not equal zero.

Because forces acting along the same axis can be either antagonistic or cooperative, **free-body diagrams** can be used to assess the balance of forces along each axis provided that the **vector components** of each force are known:

$$\begin{aligned} \sum x \text{ forces} &= \\ \sum y \text{ forces} &= \end{aligned}$$

When the sum of all forces acting on a mass along each axis of the xy -coordinate system is equal

to zero, the object is said to be in equilibrium such that no acceleration occurs:

$$\begin{aligned}\sum x \text{ forces} &= 0 \\ \sum y \text{ forces} &= 0\end{aligned}$$

In contrast to an equilibrium in which an object is moving, **static equilibrium** results when the balance of forces acting on an object maintains the absence of motion by that object.

In the prompt, a 5 kg mass is suspended in static equilibrium by a cable with a tension of 45 N. Isolating only the vertical (y-axis) forces, the magnitude of the gravitational force acting on the object can be determined by the simple equation for [weight](#):

$$F = m \cdot g = (5 \text{ kg})(9.8 \text{ m/s}^2) = \sim 50 \text{ N}$$

Because this downward (−y) gravitational force acting on the mass is greater than the 45 N tension force pulling the block upward (+y), it can be concluded that the frictional force acting on the mass must act in an opposing, upward (+y) direction to maintain the static equilibrium.

Using the equation for static equilibrium, the magnitude of the frictional force (F_f) between the mass and the wall can be calculated:

$$\begin{aligned}\sum y \text{ forces} &= \\ (F_f - 50 \text{ N} + 45 \text{ N}) &= 0 \\ F_f &= 5 \text{ N}\end{aligned}$$

Therefore, the frictional force acting between the object and the wall must be 5 N and oriented in an upward (+y) direction (**Choice A**).

Educational objective:

Static equilibrium occurs when the sum of all forces acting on a motionless object is equal to zero. Free-body diagrams are used to illustrate the directional components of translational forces.

Time spent:0 sec
Subject:
Physics

QID:402209
System:
Mechanics and Energy

A force of 50 N is used to push a wheelchair forward at constant velocity. If the wheelchair is displaced 0.5 m each second, what is the mechanical power expended when transporting the wheelchair?

- ☐ A. 13 J [1%]
✓ ☒ B. 25 W [86%]
☐ C. 100 W [10%]
☐ D. 200 J [1%]

Correct

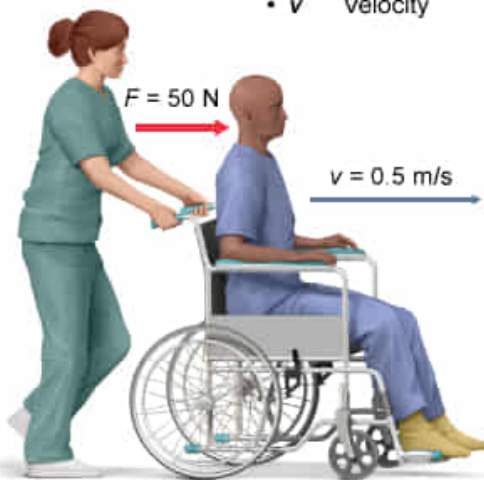
86% answered correctly

Explanation:

Power as a function of force & velocity

$$P = \frac{W}{t} = F \cdot v$$

- P Power
- W Work
- t Time
- F Force
- v Velocity



$$P = (50 \text{ N}) \cdot (0.5 \frac{\text{m}}{\text{s}}) = 25 \text{ W}$$

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In some cases, determining the total quantity of work (W) done on or by a system is sufficient for describing the transfer or expenditure of energy. However, most energy-related phenomena do not occur instantaneously. Therefore, **power (P)** is used to quantify the magnitude of energy transfer or expenditure over time (t):

$$P = \frac{W}{t}$$

The **SI unit** for power is the watt (W), where 1 W is equal to 1 joule (J) of energy expended per second (s) of time.

In the context of **translational motion**, power can be restated in terms of the change in energy per unit of time caused by the application of a **force (F)** over a distance (*d*):

$$P = \frac{(F \cdot d)}{t}$$

The above expression relating power to **mechanical work** may also be rearranged and expressed in terms of the **velocity (v)** of the object acted on by the force:

$$P = F \cdot v$$

Therefore, the power expenditure associated with the motion of an object is equal to the product of the force required to move the object and the velocity at which the object moves.

In the prompt, a 50 N force propels a wheelchair forward at a constant velocity. Because the wheelchair is displaced 0.5 m each second, the velocity of the wheelchair is equal to 0.5 m/s. Using this velocity, the power expenditure associated with the motion of the wheelchair is calculated as follows:

$$P =$$

Because 1 N·m is equal to 1 J, the power required to propel the wheelchair forward is therefore:

$$P = 25$$

Educational objective:

Power is the rate of energy transfer into or out of a system. In the context of translational motion, power is equal to the product of force and velocity.

Time spent:0 sec
Subject:
Physics

QID:402210
System:
Mechanics and Energy